

CCNA 200-301 v1.1	IP Services
Telnet Commands	
line vty 0 4 line vty 0 15	Enter vty configuration mode for the desired number of vty telnet sessions (5, or 16)
login local	Authenticate user against the local device
transport input telnet	Allow telnet connections
telnet <target device IP>	Start a telnet session from another Cisco device
CTRL+6+SHIFT, x	Toggle from the telnet session back to your local CLI
ENTER	Resume your telnet session
exit	End a telnet session
show session	Show your active outbound telnet sessions to other devices
show users	Shows active inbound telnet sessions to this device, including username and protocol
SSH Commands	
hostname	Configure a hostname for the device
ip domain-name <domain name>	Provide the device domain name
crypto key generate rsa	When prompted, choose 360 – 4096 bit key length
ip ssh version 2	Set the SSH version
username <user> privilege 15 secret <password>	Create a local user with a secret password who will have admin privilege in the session
line vty 0 15	Enter vty configuration mode for 16 vty lines
transport input ssh	Enable SSH only
transport input ssh telnet	Enable SSH and if that fails, permit telnet
login local	Require local authentication for the user
ip ssh time-out 60	The user must successfully authenticate in 60 seconds
ip ssh authentication-retries 3	The user can retry authentication 3 times before the session is terminated and they must start over
SSH Connection Commands	
ssh -l <username> <destination IP>	Start an SSH connection from one Cisco device to another
CTRL+SHIFT+6, x	Toggle back to your local CLI
ENTER	Resume your SSH session
exit logout	End your SSH session
show ssh	View active SSH sessions in our out of this device
show users	View active inbound SSH sessions to this device incl: user, protocol

HTTP Commands	
ip http server	Enable the HTTP server
ip http secure-server	Enable the HTTPS server
ip http authentication local	Set authentication to be local
username [username] privilege 15 secret [password]	Create a user that can log in
ip http port [port-number]	Set the HTTP port
ip http port 81	
ip http secure-port [secure-port-number]	Set the HTTPS port
ip http secure-port 444	
TFTP Commands	
tftp-server [file path and name]	Serve a file as a TFTP server
tftp-server flash:backup.bak	
tftp-server flash:c3560-ipservicesk9-mz.150-1.SE.bin	
copy tftp flash	Download a file from a TFTP server
copy tftp://192.168.3.47/c3560-ipservicesk9-mz.150-1.SE.bin flash:	You will be prompted for any missing information before the command is actually executed
copy flash tftp	Upload a file to a TFTP server
TFTP Verification Commands	
dir flash:	List all files in your own flash storage including their size
show flash:	
show running-config include tftp-server	View current running configuration, limiting output to lines related to TFTP

FTP Commands	
ip ftp server enable	Enable FTP server service
User <name> secret <password>	Create a non-privileged user with an encrypted password on the server
ip ftp path <directory-path> ip ftp path flash:	Specify the FTP server directory (typically flash:)
dir flash:	List all files on your local flash storage
show running-config include ftp	View current running configuration, limiting output to lines related to FTP
copy flash ftp	Initiate an FTP upload (you will be prompted for info)
copy ftp flash	Initiate an FTP download (you will be prompted for info)
ip ftp username <name> ip ftp username chrys	Pre-stage your FTP client username
ip ftp password <password> ip ftp password letmein	Pre-stage your FTP client password
ip ftp server <IP name> ip ftp server 192.168.1.10	Pre-stage the FTP server you will connect to
copy flash:<file> ftp Copy flash:startup-config ftp	Upload a file from flash: to FTP
copy ftp flash:<file>	Download a file from FTP to flash:
HTTP Client Commands	
copy http://<server>/<file> flash copy http://example.com/file.bin flash	Download a file from an HTTP server to local flash storage
copy flash:<file> http://<server>/<file> copy flash:file.bin http://example.com/	Upload a file from an HTTP server to local flash storage
Syslog Commands	
logging on	Enable logging (on by default)
logging host <syslog server> logging host 192.168.1.200	Specify the syslog server to send messages to
logging facility [local0 – local7] logging facility local2	Optionally specify the syslog facility to use (default is local7)
logging trap <minimal severity level> logging trap informational	Optionally specify the minimum severity level to send (default is informational)
service timestamps log datetime msec	Optionally specify if the message should include a timestamp to the millisecond
show logging	Display logging information including messages currently in the buffer
show logging console	Display log messages in real time at the console
terminal monitor	If you are connected via a VTY session, display log messages in real time in your session

SNMP Commands	
snmp-server community [community-string] [access-level]	Enable SNMP
snmp-server community public RO	Example: Set the community string “public” for read-only
snmp-server community private RW	Set the community string “private” for read-write
snmp-server enable traps [notification-type]	Enable a particular trap
snmp-server enable traps linkup linkdown	
Router(config)# snmp-server host <ip address> <community string> [notification-type]	Specify the SNMP manager to send the trap to
Router(config)# snmp-server host 192.168.1.100 public linkup linkdown	Include community string and notification type
show snmp	Display SNMP configuration including statistics
show running-config include snmp	Display current running configuration (SNMP only)
DHCP Commands	
ip dhcp	Enable an interface to be a DHCP client
ip helper-address <DHCP Server IP> ip helper-address 192.168.1.1	Configure an interface to be a DHCP relay
ip dhcp excluded-address <start> <end> ip dhcp excluded-address 192.168.1.1 192.168.1.10	(Global config mode) Exclude any addresses in the subnet from being offered to a DHCP client
ip dhcp pool <name> ip dhcp pool MY_DHCP_POOL	Create a DHCP pool (scope)
network <subnet ID> <subnet mask> network 192.168.1.0 255.255.255.0	(DHCP config mode) Specify the subnet ID and mask that the leased addresses will come from
default-router <gateway IP> default-router 192.168.1.1	Specify the default gateway scope option
dns-server <IP> <IP> dns-server 8.8.8.8 8.8.4.4	Specify the DNS scope option
lease <days> lease 7	Specify the DHCP lease length

DNS Commands	
ip domain-lookup	Enable DNS client / use DNS to resolve names
no ip domain-lookup	Disable DNS client / stop doing name resolution
ip name-server [dns-server-ip]	Specify the IP address of your DNS server
ip name-server 8.8.8.8	You can specify one or more DNS servers
ip domain-name [your-domain-name]	Set a default domain name that the router appends to any unqualified hostnames during DNS resolution
ip domain-name company.com	
ip host [hostname] [ip-address]	Create a static host-to-IP address mapping
ip host switch1 192.168.100.100	To be used instead of a DNS lookup for that name
ip dns server	Configure the router to be a caching DNS server; also serve ip host entry records
show hosts	Display your DNS cache and host entries
clear host	Clear all dynamically learned DNS addresses
clear host [hostname]	Clear a specific dynamically learned DNS address
ping <hostname>	See if name resolution is working (name resolution is performed before ping)
ping switch1	
ping www.cisco.com	
debug ip dns	Perform real-time debugging of DNS processes
undebug all	Stop all debugging
NTP Client and Server Commands	
ntp server [IP address hostname]	Configure the device to be an NTP client
ntp server 192.168.1.5	Specify the server to synchronize with
ntp master <1-15>	Configure the device to be an NTP server
ntp master 3	Specify the stratum level (usually 2 – 15)
NTP Peer Commands	
ntp peer <Peer IP> key <key number>	Declare an NTP peer, incl. shared secret key number
ntp peer 192.168.1.2 key 1	
ntp authenticate	Require your NTP peer to authenticate to you when synchronizing (must be done on both sides)
ntp authentication-key <key number> <hashing type> <key value>	Specify the shared secret your peer must use when authenticating to you (must be done on both sides)
ntp authentication-key 1 md5 123456	
ntp trusted-key <key number>	Mark the shared secret as trusted (do on both sides)
ntp trusted-key 1	
NTP Show Commands	
show ntp associations	Displays your server or peer IP address, the server or peer’s stratum level, your polling interval, server/peer reachability, delays and more
show ntp status	Display whether your device is synchronized by server or peer and its IP address, your own stratum level, your polling interval, your precision level, your local clock as synchronized by NTP, and more
debug ntp packets	Debug all ntp communications
undebug all	Stop all debugging